

SAKTHIMURUGAN V

Electronics and Communiation Engineer

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LinkedIn

GitHub

Portfolio

CAREER SUMMARY

Motivated Electronics and Communication Engineering student with hands-on experience in Embedded C firmware development, real-time sensor interfacing, and microcontroller design. Proficient in UART, SPI, I2C, and debugging. Seeking an entry-level Embedded Software/Firmware Engineer role.

SKILL STACK

Programming Languages	: C, C++.
Microcontrollers & Boards	: Arduino, ESP32, Raspberry Pi, STM32, 8051.
Communication Protocols	: UART, SPI, I2C, CAN.
Tools & IDEs	: Arduino IDE, Thonny IDE, STM32CubeIDE, Keil u Vision, Proteus 8, Visual Studio Code.
Embedded Skills	: RTOS, Firmware Development, Bit Manipulation.
Soft Skills	: Problem-solving, Teamwork, Adaptability, Communication, Time Management.
Languages	: Tamil, English.

EDUCATION

Bachelor of Engineering – Electronics & Communication Engineering 2022 – 2026 | CGPA: 8.5

Adhiparasakthi Engineering College, Melmaruvathur.

Higher Secondary (XII) – SRI International Matric Hr. Sec. School, Birudur – (2022) | Percentage: 85.5%

Secondary (X) – SRI International Matric Hr. Sec. School, Birudur – (2020) | Percentage: 91.4%

PROJECTS

IoT-Based Smart Agriculture System

- Developed an IoT-based smart agriculture system for real-time monitoring of soil moisture, water level, flow rate, temperature, humidity, and ambient light, with automatic irrigation and remote monitoring.
- Tools: Arduino IDE – ESP32 – Blynk IoT – OLED – Ultrasonic Sensor – Soil Moisture Sensor – DHT11 – Float Sensors – Flow Sensor – I2C Ambient Light Sensor – Relay – Solenoid Valve.

IoT-Based Smart Guide System for Visually Impaired

- Developed an IoT-based smart guide system for visually impaired individuals to assist with navigation and obstacle detection using sensors, camera-based vision, GPS tracking, and voice feedback for improved safety and independent mobility.
- Tools: Arduino IDE – ESP32 – Raspberry Pi OS – Raspberry Pi Zero 2W – Raspberry Pi Camera V2 – SIM800L – NEO-6M GPS – ToF Sensor – MPU6050.

IoT-Based Smart Patient Health Monitoring System

- Developed an IoT-based smart patient health monitoring system to continuously monitor vital health parameters and provide real-time health information, enabling early detection of abnormalities and supporting efficient remote patient monitoring.
- Tools: Blynk application – Arduino IDE – ESP32 – Heart Rate Sensor – ECG Sensor – DSB8120 – MPU6050 – OLED.

Colour Based Product Sorting Robot

- Developed an automated product sorting system using camera-based color detection and a Raspberry Pi Pico W. Classified products by color and controlled the sorting mechanism for efficient and automated operation.
- Tools: Thonny IDE – Computer Vision – Raspberry Pi Pico W – Camera – Servo Motor.

INTERNSHIP EXPERIENCE

Hardware Testing Intern – Vi Microsystems Pvt. Ltd., Chennai

Jan 2026 - Apr 2026

- Hands-on experience in embedded hardware testing, debugging, automation, and validation of embedded systems.
- Tested and programmed embedded systems using 8051, 8085, 8086, Arduino, ESP32, Raspberry Pi, and Pico.
- Validated IoT boards, embedded hardware, sensors, and peripheral interfaces for proper operation.

Embedded Systems Intern – National Small Industries Corporation, Chennai

July 2025 - Aug 2025

- Implemented sensor interfacing using ADC and GPIO for data acquisition and device control.
- Worked with UART, SPI, and I2C protocols for communication between controllers, sensors, and peripherals.
- Performed basic debugging and troubleshooting of embedded hardware, firmware, and communication issues.

CERTIFICATIONS

- Embedded Systems Design, NPTEL
- Introduction to Embedded Systems and Internet of Things, Coursera
- Embedded Systems Using C, EDUCBA
- System Design through Verilog, NPTEL